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Paleomagnetism of the Mesozoic Coast Parallel Dolerite Dikes of West Greenland

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Paleomagnetic measurements on 61 drill cores from eight sites of the Coast Parallel dikes of west Greenland (Fiskenaesset area) have yielded a pole position at 169°E, 54°N, $\alpha_{95} = 5.9°$. When these data are combined with those from the Nunarsuit-Tornarssuk area (Ketelaar 1963) the pole is 168°E, 56°N, $\alpha_{95} = 6°$. Reversely magnetized dikes are present in both areas.

The Coast Parallel dike pole lies close to the Triassic pole position for North America if corrected for Bullard's fit. The dikes were therefore intruded prior to the Mesozoic separation of Greenland and North America.

Des mesures de paléomagnétisme sur 61 carottes de forage de 8 sites de dykes parallèles à la côte ouest du Groenland (région de Fiskenaesset) ont pu établir un pôle situé à 169°E, 54°N, $\alpha_{95} = 5.9°$. En combinant ces données à celle de la région de Nunarsuit-Tornarssuk (Ketelaar 1963) on trouve un pôle à 168°E, 56°N, $\alpha = 6°$. Des dykes de magnétisation renversée sont présents dans les deux régions.

Les pôles de ces dykes ressemblent de près à ceux du Trias de l'Amérique du Nord, en les corrigeant par les courbes de Bullard. Ces dykes furent donc introduits avant la séparation du Groenland et du continent nord-américain, au Mézozoique.

[Traduit par le journal]

Paleomagnetism of the Coast Parallel Dikes of Southwest Greenland

There has been special interest in a set of relatively young basic dikes (Coast Parallel dikes) that occurs for at least 400 km along the southwest coast of Greenland. It was thought that these dikes might indicate the age of early rifting between Greenland and North America. Various features of these dikes have been reviewed recently by Watt (1969). Berthelson (1961) tentatively suggested a Tertiary age for the intrusions because they are relatively late

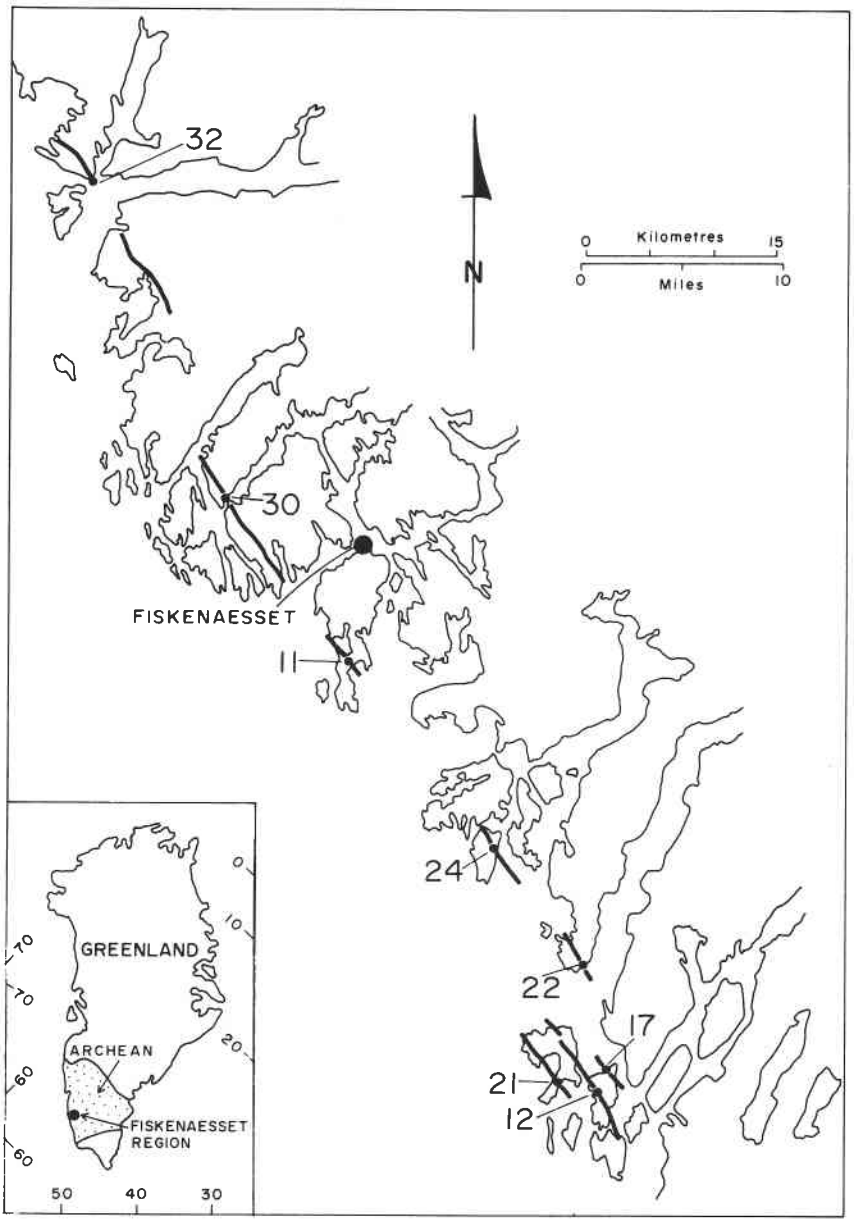


FIG. 1. Location of Coast Parallel dikes and sampling sites in the Fiskenaesset area of southwest Greenland.

and a Tertiary age was supported by preliminary paleomagnetic data. Ketelaar (1963) and Tarling (1966) reported that the dikes are reversely polarized and Tarling (1967) stated that their mean magnetization was similar in direction to that of Tertiary lavas in East Greenland. Larsen and Møller (1968) reported a K–Ar biotite age of 162 ± 5 m.y. on a lamprophyre that cuts one of the Coast Parallel dikes, and Watt (1969) reported that N. H. Gale had obtained a whole-rock K–Ar age of 138 m.y. (Jurassic) on a Coast Parallel dike.

The present study of the paleomagnetism of this swarm is part of a much larger project using different parameters (including the magnetism) of several dike swarms on both sides of

Davis Strait, in order to examine the geological fit across the Strait. Sixty-one cores were drilled and oriented with solar compass at eight sites from at least five Coast Parallel dikes in the Fiskenaesset area near the northern extremity of the swarm (Fig. 1). Sites 22, 24, 11 and 30 are from dike segments that are aligned so they may be from the same intrusion.

Two specimens were cut from each of the cores and the mean directions of the two specimens were used in final statistics to represent those cores which were homogeneously magnetized (θ, the angle between the magnetization directions of the two specimens was less than 25°) after A.F. cleaning in optimum fields. The optimum cleaning field for each site was chosen by step cleaning three test specimens from each site in fields from 50 to 800 Oe. The test specimens generally displayed good end points during the step demagnetization and optimum cleaning fields are 250 to 300 Oe. All of the specimens from four sites were cleaned in one or more additional fields to ensure that optimum A.F. peak fields had been used in the bulk cleaning.

The results of the cleaning procedures are given in Fig. 2 and Table 1. Two of the dikes, represented by site 21 and by sites 11, 22, 24, and 30 are reversely polarized and the remaining three dikes represented by sites 12, 17 and 32, are normally polarized. The presence of at least one reversal and the fact that these dikes intimately intersect older dikes of very different magnetization direction, suggests that the Coast Parallel dikes were magnetized at the time of their initial cooling. Although the number of dikes is small, the presence of a reversal indicates that they probably adequately sample paleosecular variation, so from this standpoint they provide a reliable paleopole for this part of Greenland. On the other hand, slight post consolidation tilting of the dikes could conceivably have taken place.

The Coast Parallel pole obtained from Fiskenaesset data by using present Greenland coordinates and the pole position obtained by assuming that Greenland had the pre-rifting position of Bullard *et al.* (1965) at the time of Coast Parallel intrusion, are given in Table 1.

The Bullard fit requires a rotation of about 22°. It is clear that the pre-rift pole (point B of Fig. 3) is much closer to the North American

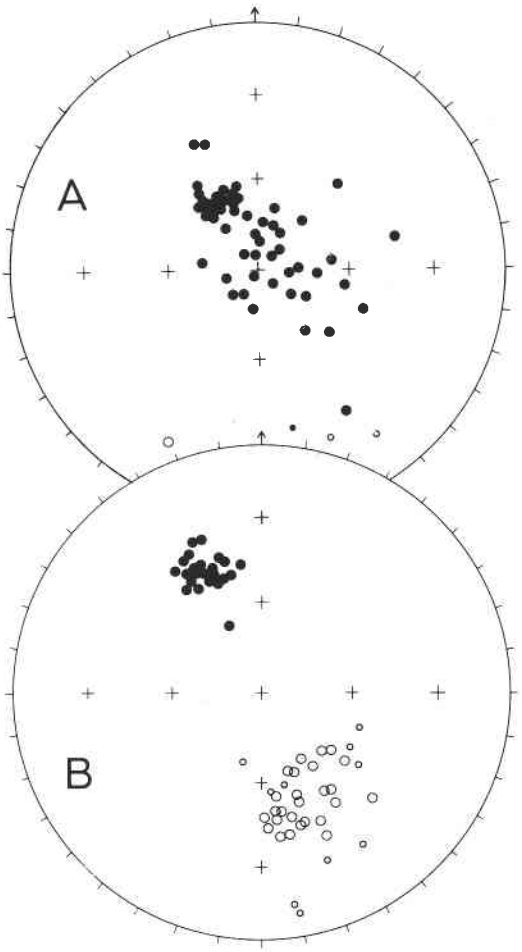


FIG. 2. Magnetization directions of Coast Parallel dikes A before, and B after, alternating field cleaning. Solid circles, north seeking down, open circles, north seeking up. Smaller circles indicate that θ, the angle between the magnetization directions of two specimens from a core exceeded 25°.

polar wandering path than is pole A, Fig. 3. This supports the concept of ocean floor spreading in this region and suggests that most or all of the spreading took place after the intrusion of these dikes. Unfortunately, the age of dike intrusion is not accurately known. The Coast Parallel pole suggests that they are Triassic.

Paleomagnetic results obtained by Tarling (1965) from dikes of probable similar age have not been combined with data from the present paper. He reported a direction of magnetization of about 335°, +70°; however, his samples were cleaned in a peak field of only 169 Oe, so they

TABLE 1. Directions of remanent magnetization before (NRM), and directions of magnetization after magnetic cleaning (cleaned); mean pole positions after magnetic cleaning (relative to present Greenland position) and with Greenland shifted relative to North America to agree with Bullard's pre-rift position

			NRM					Cleaned					N. Pole position	
Site	N	n	D	I	$\delta°$	k	af	D	I	$\delta°$	k	δ_R	Long.	Lat.
Reversely polarized sites														
11	6	6	082	80	48	3	350	162	−52	7	120	3.0	022W	−58
21	8	4	151	34	48	3	350	140	−62	5	283	2.4	018E	−60
22	8	5	141	69	15	28	300	155	−42	14	35	6.1	015W	−48
24	8	7	88	89	17	24	250	157	−48	8	102	3.0	016W	−53
30	8	6	81	84	9	68	350	154	−53	13	41	5.2	010W	−56
(A) Mean (5 Sites)			122	73	23	13		154	−52	8	90	3.8	012W	−55
(B) Mean (28 Cores)			120	77	33	6		155	−51	12	49	2.2	013W	−54
Normally polarized sites														
12	8	8	331	62	6	178	250	332	45	5	291	1.7	169E	50
17	7	7	324	58	11	53	350	330	40	7	121	2.8	170E	45
32	8	8	332	66	4	374	250	341	50	8	110	2.7	158E	55
(A) Mean (5 Sites)			329	62	4	350		334	45	6	170	3.6	165E	49
(B) Mean (23 Cores)			329	62	8	103		334	45	8	98	1.7	166E	49

Mean group pole positions after cleaning

		Number of sites	Number of cores	Long.	Lat.	d_m	d_p	
Relative to present position of Greenland	(A)	8		167°E	52°N	6.5	4.3	
	(B)		51	166°E	52°N	3.3	2.2	
	(C)	8		169°E	54°N			$\alpha_{95} = 5.9°$
Relative to North American coordinates (Bullard's pre-rift position)	(A)	8		128°E	61°N	6.5	4.3	
	(B)		51	127°E	60°N	3.3	2.2	

NOTES: N = number of cores collected at site; n = number left after excluding those that are inhomogeneously magnetized ($\theta > 25°$). D = declination, I = inclination, $\delta°$ = angular standard deviation, k = Fisher's estimate of precision, af = peak alternating cleaning field in oersteds, δ = standard error of the mean, α_{95} = half angle of cone of confidence. d_m, d_p, are semi axes of the oval of confidence.
(A) Mean pole calculated from mean site magnetization direction, irrespective of polarity.
(B) Mean pole calculated from mean core magnetization direction.
(C) Mean pole calculated from mean site pole position.

may contain a significant viscous component.

Ketelaar (1963) reported results from 23 specimens from 15 dikes between Nunarsuit and Tornarssuk. Both normally and reversely magnetized sites were discovered after partial demagnetization. Four samples were demagnetized in a peak field of 200 Oe, six in 300 Oe and thirteen in 400 Oe. His reversely magnetized dikes have directions similar to reversely magnetized dikes of the Fiskenaesset area *i.e.* to the southeast at −50°, but his normally magnetized dikes have a paleomagnetic inclination to the northwest which is about 20° more steeply inclined than that found in normally magnetized samples from the Fiskenaesset area. This might suggest the retention of a small component related to the present field direction as a result of incomplete cleaning. However, peak fields of this strength were found adequate in cleaning the Fiskenaesset samples, so we suggest the difference may be due to paleosecular variation. It is difficult to know how to mean these results with those from Fiskenaesset because only one of the 15 sites is represented by three independently oriented samples, six sites by two samples, and

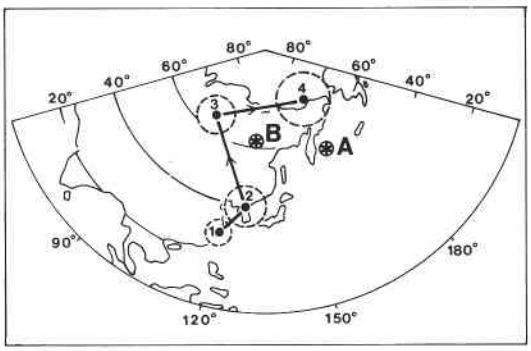


FIG. 3. Polar wandering curve of Eastern North America (Roy 1971) 1 = Upper Devonian, Lower Carboniferous; 2 = Upper Carboniferous, Permian; 3 = Triassic (Upper?); 4 = Jurassic, Cretaceous. Point A is the Coast Parallel pole using present Greenland geographic coordinates; point B is the Coast Parallel pole assuming Bullard's pre-rift fit of Greenland and North America.

eight sites by a single sample from each. For this reason all the normally magnetized specimens have been treated as a single site, and all the reversely magnetized samples as a second site. The two sites yielded magnetization directions: $N = 16$, $D = 331°$, $I = 67°$, $\alpha_{95} = 5°$ and $N = 7$, $D = 171°$, $I = -50°$, $\alpha_{95} = 11°$. When these data are combined with those from the eight Fiskenaesset sites, giving each site equal weight, the mean direction is $D = 156°$, $I = -51°$ and the combined pole position (from site pole positions, $N = 10$) is at 168°E, 56°N, $\alpha_{95} = 6°$. The combined pole position using Ketelaar's excellent earlier data and the new data from the present paper (corrected for Bullard's fit of North America and Greenland) is 127°E, 64°N.

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